

Impact of Glacial Lake Outburst Floods on Flood Frequency Statistics of the Simme River

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Abstract

Eight Glacial Lake Outburst Floods (GLOFs) from the Plaine Morte glacier (2011–2018) affected the Simme catchment before a supraglacial bypass channel ended the outburst regime in 2019. We test whether these events systematically distort flood frequency statistics using two discharge records at the Simme gauge in Lenk: daily maximum instantaneous discharge (*Tagesmaxima*, BAFU 2219, 1944–2025, $n = 82$) and daily mean discharge (GRDC 6935331, 1944–2020). Under daily maxima, GLOFs produced the annual maximum in 4 of 8 GLOF years; under daily means in 6 of 8, because their sustained outflows elevate the 24-hour average above brief storm peaks. The 2018 outburst ($\hat{T} \approx 789$ yr, daily max.) drives the statistical signal. Kolmogorov–Smirnov and Rank-Sum tests reject homogeneity with the pre-GLOF reference in both datasets (BAFU 2219: $p = 0.002$; GRDC: $p < 0.001$). After a ± 5 -day GLOF masking window, both tests fail to reject homogeneity. Naive inclusion of GLOF peaks overstates the 100-year design flood by approximately 15 %.

1 Introduction

Flood frequency analysis (FFA) rests on the assumption of stationarity: that annual maximum discharges are drawn from a single, time-invariant distribution. When a recurring process inflates extreme flows in a subset of years, this assumption is violated and fitted design quantiles become misleading.

The Simme river (canton Bern, Switzerland) provides a compelling case study. The Plaine Morte glacier hosts a supraglacial lake (Lac des Faverges) whose drainage produced eight documented Glacial Lake Outburst Floods (GLOFs) between 2011 and 2018. Since GLOFs are mechanistically distinct from the rainfall- and snowmelt-driven floods that dominate the pre-2011 record, their inclusion in a stationary FFA is conceptually problematic. Following the largest recorded GLOF in 2018, which caused approximately CHF 2.5 million in damage [GEOTEST AG, 2019], a supraglacial bypass channel was constructed in 2019 [GEOPRÆVENT AG, 2022], ending the uncontrolled outburst regime. The period 2011–2018 is therefore a closed historical episode whose statistical imprint on the long-term flood frequency record warrants retrospective investigation.

Both research questions are addressed in parallel using daily maximum instantaneous discharge (*Tagesmaxima*, BAFU 2219, 1944–2025) and daily mean discharge (GRDC 6935331, 1944–2020), making the choice of flow-aggregation statistic a key methodological variable.

2 Research Questions

1. Do recurring Glacial Lake Outburst Floods (GLOFs) from the Plaine Morte glacier systematically distort the flood frequency distribution of the Simme river at Lenk, and if so, by how much do GLOF-inclusive design quantiles exceed those from a homogeneous reference period?
2. Are the annual maximum discharges recorded during the GLOF period (2011–2018) statistically consistent with the pre-GLOF reference distribution once the direct GLOF contribution is removed, or does the post-2010 natural-flood regime show evidence of additional change?

3 Hypotheses

The research questions are operationalised as two testable statistical hypotheses.

Hypothesis 1 — GLOF inflation: GLOF-inclusive annual maxima from the 2011–2018 period are statistically incompatible with the pre-GLOF reference GEV distribution.

$H_0^{(1)}$: The 2011–2018 full annual maxima belong to the same population as the pre-GLOF reference period (no distributional shift).

$H_1^{(1)}$: The 2011–2018 full annual maxima are drawn from a stochastically larger distribution (GLOF-inflated tail).

Rejection of $H_0^{(1)}$ at $\alpha = 0.05$ confirms that the GLOF signal inflates the tail of the annual maximum series. *Tests:* KS test on Group A (GLOF-inclusive 2011–2018 peaks vs. reference GEV fitted on 1944–2010); Rank-Sum comparison 5a (GLOF-year full peaks vs. all non-GLOF years) as non-parametric corroboration.

Hypothesis 2 — Regime stationarity: Once the direct GLOF contribution is removed by masking, the 2011–2018 annual maxima are statistically indistinguishable from the pre-GLOF reference, indicating that the underlying natural-flood regime remained stationary.

$H_0^{(2)}$: The 2011–2018 GLOF-masked annual maxima belong to the same population as the pre-GLOF reference period.

$H_1^{(2)}$: The 2011–2018 GLOF-masked annual maxima differ from the pre-GLOF reference distribution (evidence of additional hydroclimatic change beyond GLOFs).

Failure to reject $H_0^{(2)}$ at $\alpha = 0.05$ supports the stationarity of the natural-flood regime and attributes the observed distributional shift entirely to the direct GLOF signal. *Tests*: KS test on Group B (GLOF-masked 2011–2018 peaks vs. reference GEV fitted on 1944–2010); Rank-Sum comparison 5c (pre-GLOF reference vs. GLOF-masked peaks) as direct non-parametric test; Rank-Sum comparison 5b (GLOF-masked peaks vs. all non-GLOF years) as additional corroboration.

4 Data

Daily maximum discharge (*Tagesmaxima*) for the Simme at Oberried/Lenk (BAFU/FOEN station 2219) spans 1974–2025. Combined with *Monatsmaxima* for 1944–1973 from the same station, this yields an extended Annual Maximum Series (AMS) of $n = 82$ years (1944–2025). The monthly and daily sub-series agree exactly over the overlap period (maximum difference = $0.000 \text{ m}^3/\text{s}$), so the two sub-series are spliced at 1974 without bias. Both records were obtained from the Swiss Federal Office for the Environment in May 2026.

The supplementary analysis uses daily mean discharge from GRDC station 6935331 (Simme at Lenk, 1944–2020, $n = 77$), providing a 67-year pre-GLOF reference.

GLOF event dates were determined from the Lac des Faverges lake-level time series provided by Ogier et al. [2021], who instrumented the lake from 2012 to 2019 at 10-minute resolution (water level, inferred volume, computed discharge). The GLOF date is defined as the timestamp of peak lake outflow: maximum discharge during the outburst phase, identified by requiring (i) lake volume to be decreasing ($\Delta V < 0$) and (ii) computed lake discharge to exceed $5 \text{ m}^3/\text{s}$. The 2011 event pre-dates the logger installation and is sourced from cantonal archive records. Uncontrolled outbursts occurred in each year from 2011 to 2018; from 2019 the bypass channel eliminates spontaneous outbursts. The eight confirmed event dates are listed in Table 1.

Table 1: Confirmed GLOF event dates, Plaine Morte glacier (2011–2018). Dates are peak lake outflow from the Lac des Faverges time series; 2011 from external sources.

Year	GLOF date
2011	27 July 2011
2012	16 July 2012
2013	2 August 2013
2014	7 August 2014
2015	1 August 2015
2016	28 August 2016
2017	19 July 2017
2018	27 July 2018

5 Methods

5.1 Series Construction and GLOF Masking

Annual block maxima are extracted from each daily discharge record. Three annual maximum series (AMS) are constructed: (i) the **full** series spanning the complete record, (ii) a **GLOF-masked** series in which values within ± 5 days of each confirmed GLOF date are set to missing before the block maximum is taken, and (iii) a **reference** series restricted to the pre-GLOF period (1944–2010). The five-day symmetric window captures the typical GLOF recession limb at this gauge. Hypothesis tests are applied exclusively to the GLOF period 2011–2018; post-2018 years are excluded because the bypass channel altered the outburst mechanism.

5.2 GEV Flood Frequency Analysis

A Generalised Extreme Value (GEV) distribution is fitted to each AMS by maximum likelihood estimation. The GEV encompasses Gumbel ($\xi = 0$), Fréchet ($\xi > 0$), and Weibull ($\xi < 0$) families, allowing the data to determine the appropriate tail behaviour. Design quantiles for $T = 2, 5, 10, 20, 50, 100, 200$ yr are derived from the fitted parameters with delta-method confidence intervals. Gringorten plotting positions are used for empirical return periods, and model fit is compared using the Akaike Information Criterion.

5.3 Homogeneity Tests

Two complementary tests compare the 2011–2018 annual maxima against the pre-GLOF reference, applied identically to both datasets.

KS test: each 2011–2018 peak is transformed to a uniform variate via the reference GEV CDF (probability integral transform). The Kolmogorov–Smirnov statistic tests departure from Uniform(0,1) with critical value $C_\alpha = K_\alpha / (\sqrt{n} + 0.12 + 0.11/\sqrt{n})$, $K_{0.05} = 1.358$ (Module 1, Eq. 3.3). Tests run separately for GLOF-inclusive (Group A, $n = 8$) and GLOF-masked (Group B, $n = 8$) peaks.

Rank-Sum test: the non-parametric Wilcoxon Rank-Sum test (Module 1, §2.4.1) compares (5a) GLOF-year full peaks vs. non-GLOF years, (5b) GLOF-masked peaks vs. non-GLOF years, and (5c) pre-GLOF reference vs. GLOF-masked peaks as a secular change check.

6 Results

6.1 Descriptive Statistics

Table 2 summarises the BAFU 2219 annual maxima. GLOF years show a mean 70% above non-GLOF years ($29.35 \text{ m}^3/\text{s}$ vs. $17.28 \text{ m}^3/\text{s}$), driven largely by the 2018 event. The relative gap is smaller in the GRDC daily-mean series (39%), consistent with daily averaging dampening brief storm peaks.

Table 2: Descriptive statistics of the annual maximum series (1944–2025, BAFU 2219 extended AMS combining Monatsmaxima 1944–1973 and Tagesmaxima 1974–2025).

	Full series	Non-GLOF years	GLOF years (2011–2018)
n	82	74	8
Mean [m ³ /s]	18.46	17.28	29.35
Std [m ³ /s]	7.51	5.55	13.45
Median [m ³ /s]	16.41	16.09	27.13
Min / Max [m ³ /s]	9.89 / 56.90	9.89 / 34.67	15.55 / 56.90

6.2 GLOF Contribution to Annual Maxima

Table 3 reports the annual maximum, its implied return period under the reference GEV, and whether the ± 5 -day masking window captures the annual maximum for each GLOF year, for both discharge records.

Table 3: Annual maxima during the GLOF period 2011–2018 under daily maximum discharge (BAFU 2219) and daily mean discharge (GRDC). The implied return period $\hat{T}$ is computed under the respective pre-GLOF reference GEV (1944–2010 for both datasets). “GLOF = max?” indicates whether the ± 5 -day masking window contains the annual maximum.

Year	GLOF date	Daily max. (BAFU 2219)			Daily mean (GRDC)		
		Q_{ann} [m ³ /s]	$\hat{T}$ [yr]	GLOF =max?	Q_{ann} [m ³ /s]	$\hat{T}$ [yr]	GLOF =max?
2011	27 Jul	27.09	17.0	No	15.08	17.4	No
2012	16 Jul	19.12	3.8	Yes	14.56	13.6	Yes
2013	2 Aug	20.59	5.0	Yes	14.99	16.6	Yes
2014	7 Aug	40.48	126.7	No	21.09	220.0	Yes
2015	1 Aug	15.55	1.9	No	10.11	1.6	Yes
2016	28 Aug	27.16	17.2	Yes	13.97	10.2	No
2017	19 Jul	27.95	19.7	No	13.94	10.0	Yes
2018	27 Jul	56.90	788.8	Yes	19.24	107.1	Yes
GLOF = ann. max (of 8 GLOF yrs)		4 / 8			6 / 8		

Under daily maxima, the GLOF produced the annual maximum in 4 of 8 years (2012, 2013, 2016, 2018); in the remaining four, storm floods dominated, including a $\hat{T} \approx 127$ -year event in 2014. Under daily means, GLOFs dominated in 6 of 8 years, because their sustained outflows elevate the 24-hour average above brief convective peaks.

6.3 GEV Parameter Estimates

Table 4 shows the MLE parameter estimates. Including GLOF years shifts $\hat{\xi}$ toward a heavier Fréchet tail in both datasets; masking restores it close to the reference value ($\Delta\hat{\xi} = +0.091$ for BAFU 2219; $+0.069$ for GRDC). The AIC is substantially lower for the masked than the full-series fit.

Table 4: GEV MLE parameter estimates and AIC for both discharge datasets.

Dataset	Series	n	$\hat{\mu}$	$\hat{\sigma}$	$\hat{\xi}$	AIC
<i>BAFU 2219 — daily maximum [m³/s]</i>						
	Full 1944–2025	82	14.94	3.95	0.251	513.66
	GLOF-masked 1944–2025	82	14.76	3.67	0.203	497.56
	Reference 1944–2010	67	14.52	3.55	0.160	399.73
<i>GRDC — daily mean [m³/s]</i>						
	Full 1944–2020	77	10.211	1.711	0.170	346.93
	GLOF-masked 1944–2020	77	10.133	1.580	0.093	328.16
	Reference 1944–2010	67	10.038	1.542	0.101	283.54

6.4 Design Quantiles

Design quantiles are compared in Tables 5 and 6. The GLOF-inclusive 100-year quantile exceeds the masked estimate by 15.0% for BAFU 2219 and 16.1% for GRDC, growing to 18–20% at 200 years. Masked and reference fits are closely aligned in both datasets.

Table 5: GEV design quantiles [m³/s] for selected return periods — BAFU 2219 extended AMS (1944–2025), with 95% delta-method confidence intervals.

T [yr]	2	5	10	20	50	100	200
Full 1944–2025	16.47	22.12	26.81	32.38	41.08	48.99	58.68
GLOF-masked	16.17	21.19	25.18	29.75	36.61	42.61	49.71
Reference 1944–2010	15.87	20.53	24.09	28.04	33.77	38.61	44.17

Table 6: GEV design quantiles [m³/s] for selected return periods — GRDC daily mean discharge.

T [yr]	2	5	10	20	50	100	200
Full 1944–2020	10.86	13.13	14.90	16.82	19.68	22.14	24.90
GLOF-masked	10.72	12.68	14.09	15.54	17.57	19.21	20.95
Reference 1944–2010	10.61	12.54	13.93	15.38	17.41	19.07	20.83

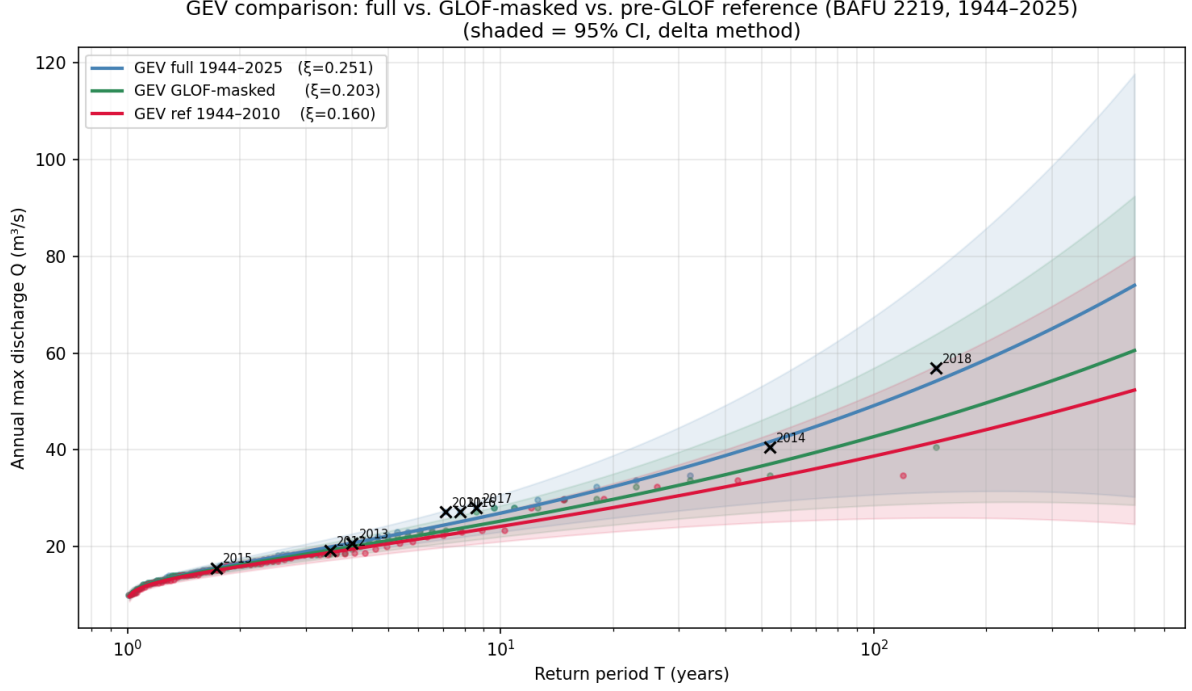


Figure 1: Return period plot for BAFU 2219 extended AMS (1944–2025): full (blue), GLOF-masked (green), and reference (red) GEV fits with 95 % delta-method confidence bands. Black crosses mark GLOF-year annual maxima at their empirical return period under the full-series fit. The 2018 event at $\hat{T} \approx 789$ yr dominates the upper tail.

6.5 Homogeneity Tests

6.5.1 KS Test

Table 7 reports the KS statistics ($n = 8$, $C_{0.05} = 0.455$). Group A (GLOF-inclusive) rejects $H_0^{(1)}$ in both datasets (BAFU 2219: $D_n = 0.610$, $p = 0.002$; GRDC: $D_n = 0.776$, $p < 0.001$). Group B (GLOF-masked) fails to reject $H_0^{(2)}$ in both ($D_n < 0.37$, $p > 0.19$).

Table 7: KS test results: 2011–2018 peaks vs. pre-GLOF reference GEV (1944–2010) for both discharge datasets ($n = 8$; $C_{0.05} = 0.455$).

Dataset	Group	D_n	$C_{0.05}$	p -value	Decision
BAFU 2219 (daily max.)	A: GLOF-inclusive	0.610	0.455	0.002	Reject $H_0^{(1)}$ at 5%
	B: GLOF-masked	0.360	0.455	0.195	Fail to reject $H_0^{(2)}$
GRDC (daily mean)	A: GLOF-inclusive	0.776	0.455	<0.001	Reject $H_0^{(1)}$ at 5%
	B: GLOF-masked	0.318	0.455	0.321	Fail to reject $H_0^{(2)}$

6.5.2 Rank-Sum Test

Table 8 shows the Rank-Sum results. Comparison 5a rejects $H_0^{(1)}$ in both datasets. Comparison 5b fails to reject $H_0^{(2)}$, confirming masking restores homogeneity. Comparison 5c (pre-GLOF reference vs. GLOF-masked peaks) also fails to reject (BAFU 2219: $p = 0.172$; GRDC: $p = 0.233$).

Table 8: Rank-Sum test results (two-sided, $\alpha = 0.05$) for both discharge datasets. Comparison 5c tests whether the GLOF-masked peaks are consistent with the pre-GLOF reference.

Dataset	Comparison	W	p -value	Decision
BAFU 2219	5a GLOF-year full vs. non-GLOF years	499.0	0.002	Reject $H_0^{(1)}$
	5b GLOF-masked vs. non-GLOF years	382.0	0.182	Fail to reject $H_0^{(2)}$
	5c Pre-GLOF ref vs. GLOF-masked peaks	188.0	0.172	Fail to reject
GRDC	5a GLOF-year full vs. non-GLOF years	486.0	0.001	Reject $H_0^{(1)}$
	5b GLOF-masked vs. non-GLOF years	349.0	0.226	Fail to reject $H_0^{(2)}$
	5c Pre-GLOF ref vs. GLOF-masked peaks	198.0	0.233	Fail to reject

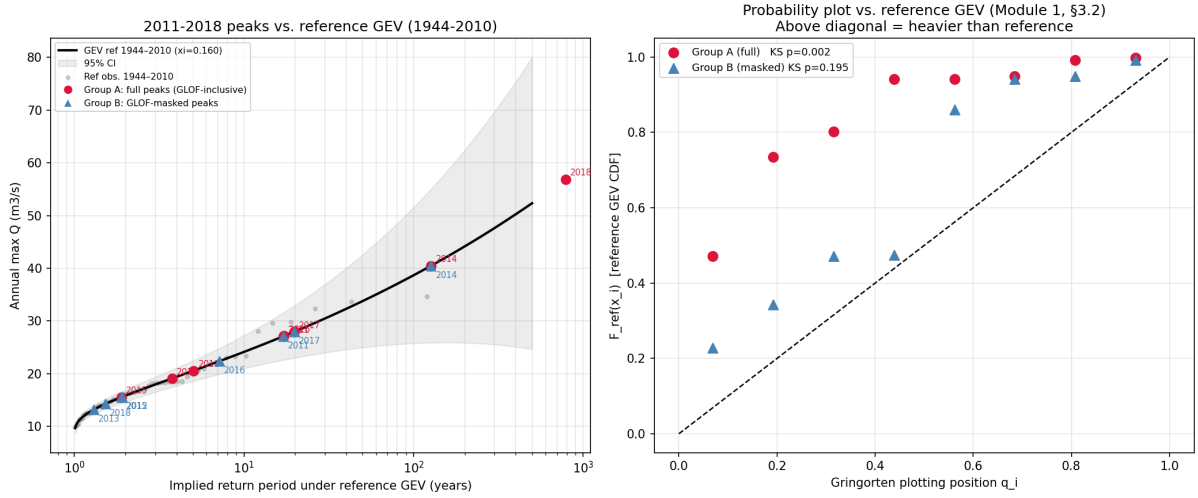


Figure 2: Left: 2011–2018 BAFU 2219 daily-maximum peaks on the reference GEV return-period curve (1944–2010). The 2018 GLOF-inclusive peak (crimson, $\hat{T} \approx 789$ yr) dominates; after masking it drops to $\hat{T} \approx 1.5$ yr (blue triangle). Right: probability plot (Module 1, §3.2). Group A lies above the diagonal almost entirely owing to 2018; Group B is broadly consistent with the 1:1 line.

7 Discussion

GLOF contribution to the AMS. Under daily maximum discharge, the outburst produced the annual maximum in 4 of 8 GLOF years (2012, 2013, 2016, 2018). In the remaining four (2011, 2014, 2015, 2017), independent storm floods recorded higher instantaneous peaks, including a $\hat{T} \approx 127$ -year rainfall event in 2014 ($Q = 40.48 \text{ m}^3/\text{s}$) and three approximately 17–20-year events. Under daily mean discharge, GLOFs dominated in 6 of 8 years, because their sustained multi-hour outflows elevate the 24-hour average above brief convective storm peaks. The choice of aggregation statistic therefore fundamentally shapes the apparent GLOF influence on the AMS: daily maxima are the more relevant basis for short-duration hydraulic design, while daily means better characterise flood volume relevant to inundation duration or reservoir routing.

Statistical signal driven by the 2018 event. The 2018 outburst ($Q = 56.90 \text{ m}^3/\text{s}$, $\hat{T} \approx 789$ yr under the reference GEV) is the primary reason Group A fails the KS test. A nearly 800-year event in an 8-year window is statistically improbable but not impossible ($p \approx 1\%$). The causal attribution to the GLOF rests on the masking experiment: after the ± 5 -day window,

2018’s annual maximum collapses from $56.90 \text{ m}^3/\text{s}$ to $14.28 \text{ m}^3/\text{s}$ ($\hat{T} \approx 1.5 \text{ yr}$), confirming that the extreme peak was attributable to the outburst, not concurrent meteorological forcing. With only $n = 8$ test observations and a critical value $C_{0.05} = 0.455$ that is itself sensitive to a single outlier, the formal KS test carries limited inferential weight in isolation; the masking experiment provides more compelling mechanistic evidence.

Hypothesis 1 confirmed. Both tests reject $H_0^{(1)}$ for Group A in both datasets (BAFU 2219: KS $p = 0.002$, Rank-Sum $p = 0.002$; GRDC: KS $p < 0.001$, Rank-Sum $p = 0.001$). The stronger rejection for GRDC is consistent with GLOFs dominating 6 of 8 years under daily means versus 4 of 8 under daily maxima. The GLOF-inclusive GEV overstates the 100-year design flood by 15.0 % for BAFU 2219 and 16.1 % for GRDC, driven by an upward shift in shape parameter ($\hat{\xi} : 0.160 \rightarrow 0.251$ for BAFU 2219). The inflation is dominated by the 2018 event rather than being a systematic feature of all eight GLOFs.

Hypothesis 2 supported. After masking, neither test rejects $H_0^{(2)}$ in either dataset (BAFU 2219: KS $p = 0.195$, Rank-Sum $p = 0.182$; GRDC: KS $p = 0.321$, Rank-Sum $p = 0.226$). The secular-change check (5c: pre-GLOF reference vs. GLOF-masked peaks) also fails to reject in both (BAFU 2219: $p = 0.172$; GRDC: $p = 0.233$), confirming that once GLOF contributions are removed, the natural-flood regime during 2011–2018 is statistically indistinguishable from the pre-GLOF reference. The convergence of results across two different discharge statistics strengthens this conclusion.

Implications of the bypass intervention. The bypass channel constructed in 2019 [GEOTEST AG, 2019] has ended the uncontrolled GLOF episode. The GLOF-masked or pre-GLOF reference GEV is now the appropriate design distribution for natural flood hazard at Lenk. The 15–18 % overestimation from naive inclusion of unmasked peaks would lead to unnecessarily conservative standards in any post-bypass hazard reassessment.

Limitations. The principal limitation is sample size: only $n = 8$ GLOF-period years constrain the test groups, making both tests sensitive to a single outlier. Failure to reject Hypothesis 2 cannot be interpreted as confirmation of stationarity. Future work should extend the analysis as post-bypass data accumulate, apply mixture models that explicitly separate the GLOF and natural-flood populations [Kite, 1977], and monitor how lake drainage dynamics evolve as Plaine Morte continues to lose mass.

8 Conclusion

This study examines the influence of eight Plaine Morte GLOFs (2011–2018) on flood frequency statistics at the Simme gauge in Lenk, using daily maximum (BAFU 2219, 1944–2025, $n = 82$) and daily mean discharge (GRDC 6935331, 1944–2020). The key findings are:

1. **GLOF dominance depends critically on the discharge statistic.** Under daily maxima (BAFU 2219), the GLOF produced the annual maximum in 4 of 8 GLOF years; under daily means (GRDC), in 6 of 8, because sustained outflows elevate the daily average above brief storm peaks. The GLOF period also included an approximately 127-year natural storm flood (2014) and three 17–20-year events (2011, 2015, 2017).
2. **One extreme event dominates the signal.** The 2018 outburst ($Q = 56.90 \text{ m}^3/\text{s}$, $\hat{T} \approx 789 \text{ yr}$) drives the KS rejection. After masking, its return period collapses to $\hat{T} \approx 1.5 \text{ yr}$, confirming the GLOF as the mechanistic cause.

3. **Tests reject homogeneity for GLOF-inclusive peaks.** KS and Rank-Sum tests reject $H_0^{(1)}$ for Group A in both datasets (BAFU 2219: $p = 0.002$; GRDC: $p < 0.001$). With only $n = 8$ test observations, results are sensitive to a single extreme event.
4. **Masking restores homogeneity.** After the ± 5 -day mask, neither test rejects $H_0^{(2)}$ in either dataset (BAFU 2219: KS $p = 0.195$, Rank-Sum $p = 0.182$; GRDC: KS $p = 0.321$, Rank-Sum $p = 0.226$). The secular-change check (5c) also fails to reject (BAFU 2219: $p = 0.172$; GRDC: $p = 0.233$), confirming the natural-flood regime is statistically unchanged across the two periods.
5. **Design flood inflation is substantial but event-driven.** Unmasked GEV fitting overstates the 100-year design quantile by approximately 15–16 % in both datasets, driven by the upward shift in shape parameter ($\hat{\xi} : 0.160 \rightarrow 0.251$ for BAFU 2219).

For engineering applications, the GLOF-masked or pre-GLOF reference GEV is recommended as the design distribution for natural flood hazard at Lenk. The 2019 bypass channel makes this recommendation directly applicable to current planning.

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